Spatial Statistics - STAT 669 - Homework 7.3

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1. 8.17

Solution: The two covariance functions are

$$K_0(t) = e^{-|t|}, \qquad K_1(t) = (1 - |t|)_+,$$

where $(x)_+ = \max\{x, 0\}$, and P_j denotes the law of a mean-zero stationary Gaussian process on $[0, L]$ with covariance K_j, for $j = 0, 1$. Take $L > 1$ and $\delta_n = (L-1)/n$, and define

$$S_n = \sum_{j=1}^{n} \big\{Z(\delta_n j) - Z(\delta_n(j-1))\big\}\big\{Z(1 + \delta_n j) - Z(1 + \delta_n(j-1))\big\}.$$

Set $X_{j,n} = Z(\delta_n j) - Z(\delta_n(j-1))$ and $Y_{j,n} = Z(1+\delta_n j) - Z(1 + \delta_n(j-1))$, so $S_n = \sum_{j=1}^{n} X_{j,n} Y_{j,n}$. Since $EZ = 0$,

$$E_j S_n = \sum_{k=1}^{n} \mathrm{Cov}_j(X_{k,n}, Y_{k,n}).$$

By stationarity $\mathrm{Cov}_j(X_{k,n}, Y_{k,n})$ is the same for every k, and expanding it in terms of K_j gives

$$\begin{aligned}\mathrm{Cov}_j(X_{k,n}, Y_{k,n}) &= E_j\big[(Z(\delta_n k) - Z(\delta_n(k-1)))(Z(1+\delta_n k) - Z(1+\delta_n(k-1)))\big] \\ &= 2K_j(1) - 2K_j(1+\delta_n),\end{aligned}$$

hence $E_j S_n = 2n\{K_j(1) - K_j(1+\delta_n)\}$.

For $K_0(t) = e^{-|t|}$, a direct computation gives

$$K_0(1+\delta) - K_0(1) = e^{-(1+\delta)} - e^{-1} = e^{-1}(e^{-\delta} - 1) = -e^{-1}\delta + O(\delta^2),$$

so $K_0(1) - K_0(1+\delta_n) = e^{-1}\delta_n + O(\delta_n^2)$ and

$$E_0 S_n = 2n\big(e^{-1}\delta_n + O(\delta_n^2)\big) = 2e^{-1}(L-1) + O\Big(\tfrac{(L-1)^2}{n}\Big) \;\longrightarrow\; 2e^{-1}(L-1).$$

For $K_1(t) = (1 - |t|)_+$, since $\delta_n < 1$ we have $K_1(1) = 0$ and $K_1(1+\delta_n) = 0$, so $E_1 S_n = 0$ for all n.

Since $2e^{-1}(L-1) > 0$, the two sequences have distinct limits. It remains to show the variances vanish so that S_n concentrates around each limit.

Both covariance functions satisfy $K_j(0) - K_j(\delta) = |\delta| + O(\delta^2)$ as $\delta \to 0$, and since $X_{k,n}$ and $Y_{k,n}$ are both increments of the same stationary process over an interval of length δ_n,

$$\sigma_{j,n}^2 := \mathrm{Var}_j(X_{k,n}) = \mathrm{Var}_j(Y_{k,n}) = 2\{K_j(0) - K_j(\delta_n)\} = 2\delta_n + O(\delta_n^2).$$

Write

$$\operatorname{Var}_j S_n = \sum_{k=1}^{n} \operatorname{Var}_j(X_{k,n}Y_{k,n}) + \sum_{k\neq\ell} \operatorname{Cov}_j(X_{k,n}Y_{k,n}, X_{\ell,n}Y_{\ell,n}).$$

Since Z is Gaussian, Isserlis' theorem expresses every fourth-order joint moment as a sum of products of second-order covariances. For the diagonal terms: $\operatorname{Var}_j(X_{k,n}Y_{k,n})$ depends only on $\operatorname{Var}_j(X_{k,n})$, $\operatorname{Var}_j(Y_{k,n})$, and $\operatorname{Cov}_j(X_{k,n}, Y_{k,n})$, each of order δ_n, so $\operatorname{Var}_j(X_{k,n}Y_{k,n}) = O(\delta_n^2)$. For $k \neq \ell$: every term in the Isserlis expansion of $\operatorname{Cov}_j(X_{k,n}Y_{k,n}, X_{\ell,n}Y_{\ell,n})$ contains at least one cross-index covariance, e.g. $\operatorname{Cov}_j(X_{k,n}, X_{\ell,n})$. The lag between $X_{k,n}$ and $X_{\ell,n}$ is $\delta_n|k-\ell| \geq \delta_n > 0$; explicitly,

$$\operatorname{Cov}_j(X_{k,n}, X_{\ell,n}) = K_j\big(\delta_n(k-\ell+1)\big) - K_j\big(\delta_n(k-\ell)\big) - K_j\big(\delta_n(k-\ell)\big) + K_j\big(\delta_n(k-\ell-1)\big),$$

a second difference of K_j at a nonzero lag, which is $O(\delta_n^2)$. Hence each off-diagonal covariance is $O(\delta_n^3)$, giving

$$\operatorname{Var}_j S_n = O(n\delta_n^2) + O(n^2\delta_n^3) = O\left(\frac{(L-1)^2}{n}\right) + O\left(\frac{(L-1)^3}{n}\right) \longrightarrow 0.$$

By Chebyshev's inequality, $S_n \xrightarrow{P_0} 2e^{-1}(L-1)$ and $S_n \xrightarrow{P_1} 0$ in probability. Setting $a = e^{-1}(L-1)$, which lies strictly between 0 and $2e^{-1}(L-1)$, the sets $A_n = \{S_n > a\}$ satisfy $P_0(A_n) \to 1$ and $P_1(A_n) \to 0$, so $P_0 \perp P_1$.

2. 8.18

Solution: The Matérn spectral density in $\mathbb{R}^d$ is

$$f_j(\omega) = \frac{\phi_j}{\left(\alpha_j^2 + \frac{1}{2\kappa_j}\,\omega^T A_j \omega\right)^{\kappa_j + d/2}},$$

with $\phi_j > 0$, $\alpha_j > 0$, $\kappa_j > 0$, and A_j symmetric with $\operatorname{tr}(A_j) = 0$. We determine when

$$\int_{|\omega|>R} \left(1 - \frac{f_1(\omega)}{f_0(\omega)}\right)^2 d\omega < \infty$$

holds for some $R < \infty$.

As $|\omega| \to \infty$ the α_j^2 terms are negligible compared to the quadratic form, so

$$f_j(\omega) \sim \phi_j \left(\tfrac{1}{2\kappa_j}\omega^T A_j \omega\right)^{-(\kappa_j + d/2)}.$$

Suppose first that at least one parameter other than α differs.

If $A_1 \neq A_0$, since A_1 and A_0 are distinct positive-definite matrices there exists a direction $\theta_0 \in S^{d-1}$ such that $\theta_0^T A_1 \theta_0 \neq \theta_0^T A_0 \theta_0$. Along the ray $\omega = r\theta_0$,

$$\frac{f_1(\omega)}{f_0(\omega)} \sim \frac{\phi_1 (2\kappa_1)^{\kappa_1 + d/2}}{\phi_0 (2\kappa_0)^{\kappa_0 + d/2}} \cdot \frac{(\theta_0^T A_0 \theta_0)^{\kappa_0 + d/2}}{(\theta_0^T A_1 \theta_1)^{\kappa_1 + d/2}} \cdot r^{2(\kappa_0 - \kappa_1)},$$

which converges to a limit $C(\theta_0) \neq 1$ when $\kappa_0 = \kappa_1$, or diverges/vanishes otherwise. In either case $(1 - f_1/f_0)^2$ does not tend to 0 along $\omega = r\theta_0$, so

$$\int_{|\omega|>R} \Big(1 - \frac{f_1}{f_0}\Big)^2 d\omega \geq \int_R^\infty \int_{B(\theta_0,\varepsilon)} \Big(1 - \frac{f_1}{f_0}\Big)^2 r^{d-1}\, d\sigma(\theta)\, dr = \infty$$

for any small ball $B(\theta_0, \varepsilon) \subset S^{d-1}$.

If $A_1 = A_0 = A$ but $\kappa_1 \neq \kappa_0$, write $Q(\omega) = \omega^T A \omega$ and $r = |\omega|$. Then $f_1(\omega)/f_0(\omega) \sim C\, r^{-2(\kappa_1 - \kappa_0)}$. For $\kappa_1 > \kappa_0$ the ratio tends to 0, so $(1 - f_1/f_0)^2 \to 1$ and $\int_{|\omega|>R} (\cdots)\, d\omega \geq c \int_R^\infty r^{d-1} dr = \infty$. For $\kappa_1 < \kappa_0$ the ratio grows like $r^{2(\kappa_0 - \kappa_1)}$, so $(1 - f_1/f_0)^2 \sim C^2 r^{4(\kappa_0 - \kappa_1)} \to \infty$, and the integral diverges even faster.

If $A_1 = A_0$, $\kappa_1 = \kappa_0 = \kappa$, but $\phi_1 \neq \phi_0$, then $f_1/f_0 \to \phi_1/\phi_0 \neq 1$ uniformly in ω, so $(1 - f_1/f_0)^2 \to (1 - \phi_1/\phi_0)^2 > 0$ and (8.16) fails.

So (8.16) requires all parameters except possibly α to be equal.

Now take $\phi_1 = \phi_0 = \phi$, $\kappa_1 = \kappa_0 = \kappa$, $A_1 = A_0 = A$, but $\alpha_1 \neq \alpha_0$. Then

$$1 - \frac{f_1(\omega)}{f_0(\omega)} = 1 - \left(\frac{\alpha_0^2 + \frac{1}{2\kappa} Q(\omega)}{\alpha_1^2 + \frac{1}{2\kappa} Q(\omega)} \right)^{\kappa + d/2}.$$

Set $u = \frac{1}{2\kappa} Q(\omega)$; for large $|\omega|$,

$$\frac{\alpha_0^2 + u}{\alpha_1^2 + u} = 1 + \frac{\alpha_0^2 - \alpha_1^2}{\alpha_1^2 + u} = 1 + O(u^{-1}),$$

and $(1+x)^s = 1 + sx + O(x^2)$ for small x gives

$$1 - \frac{f_1(\omega)}{f_0(\omega)} = -(\kappa + d/2) \cdot \frac{\alpha_0^2 - \alpha_1^2}{\alpha_1^2 + u} + O(u^{-2}) = O(u^{-1}) = O\big(|\omega|^{-2}\big).$$

So $(1 - f_1/f_0)^2 = O(|\omega|^{-4})$, and in polar coordinates

$$\int_{|\omega|>R} \Big(1 - \frac{f_1}{f_0}\Big)^2 d\omega \;\asymp\; \int_R^\infty r^{d-1} \cdot r^{-4}\, dr = \int_R^\infty r^{d-5}\, dr,$$

which converges if and only if $d \leq 3$.

For $d \leq 3$, (8.16) holds for two Matérn spectral densities if and only if all parameters other than α agree. For $d \geq 4$, changing only α already makes the integral diverge, so (8.16) holds only when all parameters $(\phi, \alpha, \kappa, A)$ are the same.